

[EECS 304] Control I

Sheet (5) Routh Stability Criterion

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Overview

We can check the stability of the system in S-plane using Routh stability criterion:

- 1 Get characteristic equation of the system:

$$F(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0 = 0$$

- 2 All coefficients must be positive.
- 3 Make the following table, and check that all the terms in the first column of the array have positive signs.

S^n	a_n	a_{n-2}	...	a_0
S^{n-1}	a_{n-1}	a_{n-3}	...	
S^{n-2}	$b_{n-1} = \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}}$	$b_{n-1} = \frac{a_{n-1}a_{n-4} - a_n a_{n-5}}{a_{n-1}}$		
...	...	...		
S^1	f_1			
S^0	g_1			

Notes:

- For 2^{nd} order system, only step (2) is sufficient.
- For 3^{rd} order system, only step (2) and $(a_1 a_2 > a_0 a_3)$ are sufficient.
- For 4^{th} or higher order, use steps (2) and (3).
- The number of changes in the sign of the coefficients in the 1^{st} column equals the number of roots of the characteristic equation with positive real parts.

Problem 1

Problem 1

Test the stability for the following polynomials; hence determine whether any of the roots are in the RHP:

① $F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$

② $F(s) = s^5 + 5s^4 + 11s^3 + 23s^2 + 28s + 12$

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		
s^4		
s^3		
s^2		
s^1		
s^0		

Problem 1

a)

$$F(s) = 1s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5	$ $	1
s^4		
s^3		
s^2		
s^1		
s^0		

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		1
s^4		3
s^3		
s^2		
s^1		
s^0		

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		1	2
s^4		3	
s^3			
s^2			
s^1			
s^0			

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		1	2
s^4		3	6
s^3			
s^2			
s^1			
s^0			

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		1	2	6
s^4		3	6	
s^3				
s^2				
s^1				
s^0				

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		1	2	6
s^4		3	6	9
s^3				
s^2				
s^1				
s^0				

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		1	2	6
s^4		3	6	9
s^3		ϵ		
s^2				
s^1				
s^0				

$$b_1 = \frac{3 * 2 - 1 * 6}{3} = 0 = \epsilon$$

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		1	2	6
s^4		3	6	9
s^3		€	3	
s^2				
s^1				
s^0				

$$b_2 = \frac{3 * 6 - 1 * 9}{3} = 3$$

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		1	2	6
s^4		3	6	9
s^3		ϵ	3	
s^2		$\epsilon * 6 - 3 * 3$		
s^1		ϵ		
s^0				

$$c_1 = \frac{\epsilon * 6 - 3 * 3}{\epsilon}$$

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		1	2	6
s^4		3	6	9
s^3		ϵ	3	
s^2		$\frac{\epsilon * 6 - 3 * 3}{\epsilon}$	9	
s^1				
s^0				

$$c_2 = \frac{\epsilon * 9 - 3 * 0}{\epsilon} = 9$$

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5		1	2	6
s^4		3	6	9
s^3		ϵ	3	
s^2		$\epsilon * 6 - 3 * 3$	9	
		ϵ		
s^1		$3 - \frac{9\epsilon^2}{6\epsilon - 9}$		
s^0				

$$d_1 = 3 - \frac{9\epsilon^2}{6\epsilon - 9}$$

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5	1	2	6
s^4	3	6	9
s^3	ϵ	3	
s^2	$\epsilon * 6 - 3 * 3$ 	9	
s^1	$3 - \frac{\epsilon}{9\epsilon^2}$		
s^0	$6\epsilon - 9$ 9		

$$e_1 = 9$$

Problem 1

a)

$$F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$$

s^5	1	2	6
s^4	3	6	9
s^3	ϵ	3	
s^2	$\frac{6\epsilon - 9}{\epsilon}$	9	
s^1	$3 - \frac{9\epsilon^2}{6\epsilon - 9}$		
s^0	9		

Let $\epsilon = 0.01$

s^5	+
s^4	+
s^3	+
s^2	-
s^1	+
s^0	+

$\therefore$ no. of sign changes = 2

$\therefore$ no. of poles at RHP = 2

Problem 1

b)

$$F(s) = s^5 + 5s^4 + 11s^3 + 23s^2 + 28s + 12$$

s^5	1	11	28
s^4	5	23	12
s^3	6.4	25.6	
s^2	3	12	
s^1	0	0	
s^0			

Problem 1

b)

$$F(s) = s^5 + 5s^4 + 11s^3 + 23s^2 + 28s + 12$$

s^5	1	11	28
s^4	5	23	12
s^3	6.4	25.6	
s^2	3	12	
s^1	0	0	
s^0			

Use auxiliary equation: $3s^2 + 12 = 0$

Take $d/ds \rightarrow 6s$

Problem 1

b)

$$F(s) = s^5 + 5s^4 + 11s^3 + 23s^2 + 28s + 12$$

s^5	1	11	28
s^4	5	23	12
s^3	6.4	25.6	
s^2	3	12	
s^1	6	0	
s^0			

Use auxiliary equation: $3s^2 + 12 = 0$

Take $d/ds \rightarrow 6s$

Problem 1

b)

$$F(s) = s^5 + 5s^4 + 11s^3 + 23s^2 + 28s + 12$$

s^5	1	11	28
s^4	5	23	12
s^3	6.4	25.6	
s^2	3	12	
s^1	6	0	
s^0	12		

Problem 1

b)

$$F(s) = s^5 + 5s^4 + 11s^3 + 23s^2 + 28s + 12$$

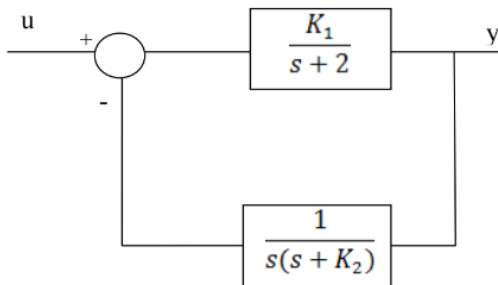
s^5	1	11	28
s^4	5	23	12
s^3	6.4	25.6	
s^2	3	12	
s^1	0	0	
s^0	12		

$\therefore$ no. of sign changes = 0 and I have 0 element
 $\therefore$ system is critically stable.

Problem 2

Problem 2

For the system shown in figure, find the conditions on K_1 and K_2 to make the system stable. Plot the region of stability for K_1 and K_2 .



Problem 2

$$TF = \frac{K_1 s(s + K_2)}{s(s + K_2)(s + 2) + K_1}$$

$\therefore$ c/ch. eq. $\rightarrow 1 + GH = 0$

$$\therefore s(s + K_2)(s + 2) + K_1 = 0$$

$$s^3 + (K_2 + 2)s^2 + 2K_2s + K_1 = 0$$

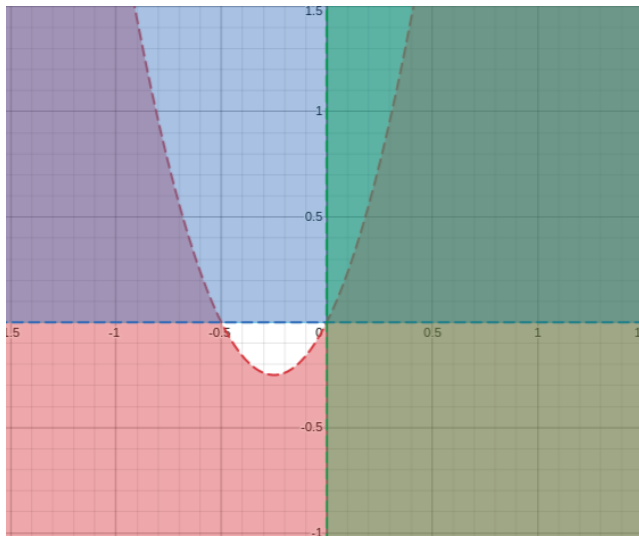
- For stability $\rightarrow K_1 > 0, K_2 + 2 > 0 \rightarrow K_2 > -2, K_2 > 0$
- For 3rd system $\rightarrow a_1 a_2 > a_0 a_3$

$$\therefore 2K_2^2 + 4K_2 > K_1$$

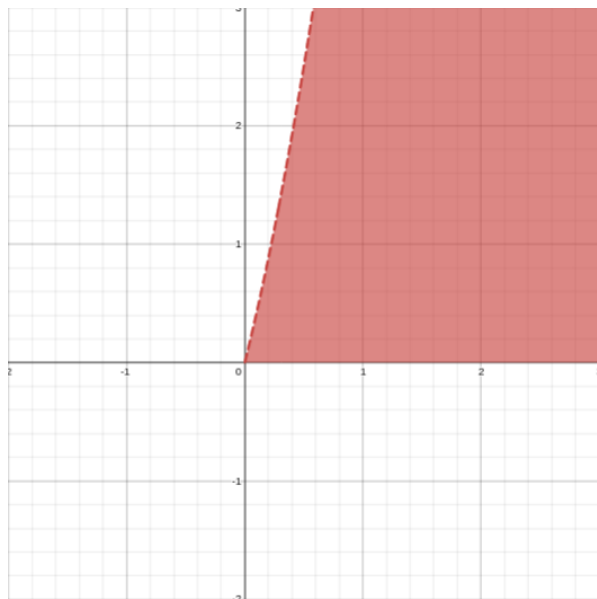
Therefore, complete condition will be:

- $K_2 > 0$
- $0 < K_1 < 2K_2^2 + 4K_2$

Problem 2



Problem 2



Problem 3

Problem 3

A unity feedback control system has the forward transfer function:

$$G_p(s) = \frac{K_v}{s(4s + 1)(s + 1)}$$

- 1 The steady state value of the error is desired to be less than or equal to 0.1 for a reference input $r(t) = 1 + t$. Determine the minimum value of K_v that satisfies this requirement.
- 2 Check the stability of the system for the value of K_v obtained in part (1) and comment on your result.

Problem 3

a)

$$\because r(t) = 1 + t \rightarrow R(s) = \frac{1}{s} + \frac{1}{s^2}$$

$$\therefore e_{ss}|_{total} = \frac{M_1}{1 + K_p} + \frac{M_2}{K_v}$$

$$M_1 = 1, \quad M_2 = 1,$$

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{K_v}{s(4s + 1)(s + 1)} = \infty$$

$$K_v = \lim_{s \rightarrow 0} sG(s)H(s) = \lim_{s \rightarrow 0} \frac{K_v}{(4s + 1)(s + 1)} = K_v$$

$$\therefore e_{ss}|_{total} = 0 + \frac{1}{K_v} \leq 0.1$$

$$\therefore K_v|_{min} = 10$$

Problem 3

b) $\therefore$ c/ch. eq. $\rightarrow 1 + GH = 0$

$$\therefore s(4s + 1)(s + 1) + K_v = 0$$

$$\therefore 4s^3 + 5s^2 + s + K_v = 0$$

- For stability $\rightarrow K_v > 0$
- For 3rd system $\rightarrow a_1 a_2 > a_0 a_3$

$$\therefore 5 > 4K_v$$

Therefore, complete condition will be:

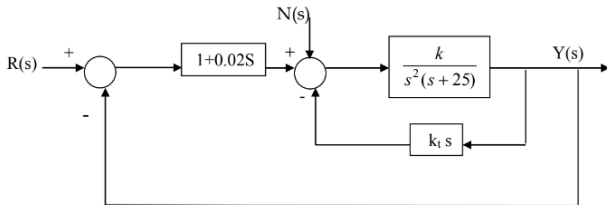
$$0 < K_v < 1.25$$

Therefore, $K_v = 10$ can't be used as the system will be unstable, so there will be a trade-off between the system's stability and its accuracy.

Problem 4

Problem 4

The block diagram of a feedback control system is shown in the figure. The error signal is defined to be $e(t)$. Find the steady state error of the system in terms of K and K_t when the input is a unit ramp function. Give the constraints on the values of k and kt so that the answer is valid. Let $n(t) = 0$ for this part.



Problem 4

$$TF = \frac{G}{1 + GH}, \quad G = G_1 G_2$$

$$G_1 = 1 + 0.02s, \quad G_2 = \frac{K}{s^3 + 25s^2 + K_t K s}$$

$$\therefore TF = \frac{K(1 + 0.02s)}{s^3 + 25s^2 + K_t K s + K(1 + 0.02s)}$$

For Ramp input $\rightarrow e_{ss} = \frac{1}{K_v}$

$$\therefore K_v = \lim_{s \rightarrow 0} sG(s)H(s) = \lim_{s \rightarrow 0} s \frac{K(1 + 0.02s)}{s(s^2 + 25s + K_t K)} = \frac{1}{K_t}$$

$$\therefore e_{ss} = K_t$$

Problem 4

The constraints on $K, K_t \rightarrow$ the system must be **stable**.

$$\therefore \text{c/ch. eq.} \rightarrow s^3 + 25s^2 + (K_t K + 0.02K)s + K = 0$$

To be stable $\rightarrow K > 0, K_t K + 0.02K > 0 \rightarrow K_t > 0.02$

$$25(K_t K + 0.02K) > K \rightarrow K_t > 0.02$$

$$\therefore K > 0, K_t > 0.02$$

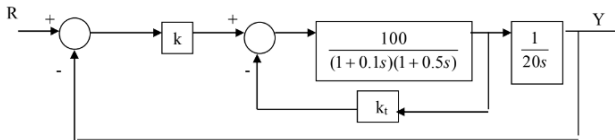
Problem 5

Problem 5

The block diagram of a control system is shown the following figure. Find the step, ramp and parabolic error constants. The error signal is defined to be $e(t)$. Find the steady state errors of the system in terms of K and K_t , when the following inputs are applied:

- ① $r(t) = 6 + 8t$
- ② $r(t) = 2t + 7t^2$

What constraint must be made on the values of K and K_t so that the answers are valid? Determine the minimum steady state error that can be achieved with a unit ramp input.



Problem 5

$$G(s)H(s) = \frac{100K}{s[(s+10)(s+2) + 2000K_t]}$$

a)

$$K_p = \lim_{s \rightarrow 0} G_p(s)H(s) = \infty$$

$$K_v = \lim_{s \rightarrow 0} sG_p(s)H(s) = \frac{100K}{20 + 2000K_t} = \frac{5K}{1 + 100K_t}$$

$$K_a = \lim_{s \rightarrow 0} s^2 G_p(s)H(s) = 0$$

b)

- For $r(t) = 6 + 8t \rightarrow e_{ss} = \frac{6}{1 + K_p} + \frac{8}{K_v} = \frac{8}{\frac{5K}{1 + 100K_t}}$

- For $r(t) = 2t + 7t^2 \rightarrow e_{ss} = \frac{2}{K_v} + \frac{14}{K_a} = \infty$

Problem 5

c) The system must be stable

$$\therefore s^3 + 12s^2 + (20 + 2000K_t)s + 100K = 0$$

$$\therefore K > 0$$

$$\therefore 20 + 2000K_t > 0 \rightarrow K_t > -0.01$$

$$\therefore 12(20 + 2000K_t) > 100K \rightarrow \frac{5K}{1 + 100K_t} < 12$$

d)

$$e_{ss} = \frac{1}{K_v} = \frac{1 + 100K_t}{5K}$$

$$\therefore e_{ss}|_{min} = \frac{1}{\frac{5K}{1 + 100K_t}} = \frac{1}{12}$$

Thank you!

Questions?

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